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Improving System Performance

Computer Engineering 1

Agenda

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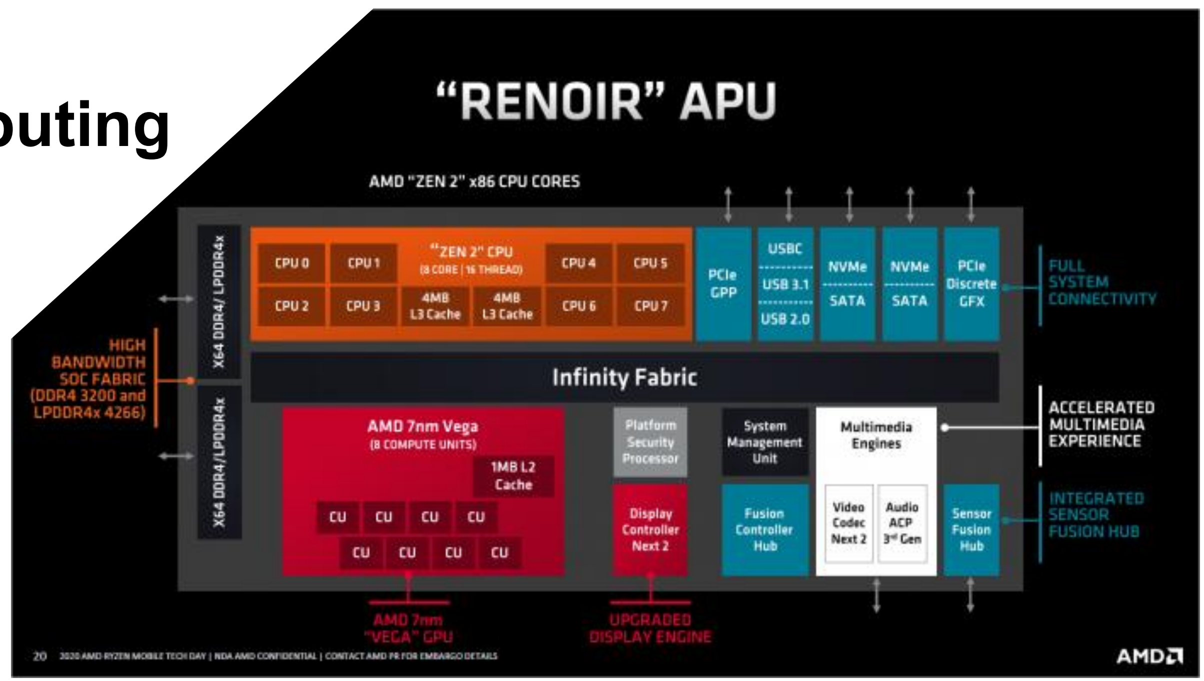
- Motivation
- Aspects of Optimization
- Bus Architectures
- Instruction Set Architectures (ISA)
- Pipelining
- Parallel Computing



Motivation

- **Intel Pentium E5800**
  - 3.2 GHz
  - 2 Cores
  - 2 MB cache
  - Released Q4 2010
  - **PassMark: 1184**
  - **TDP: 65 W**

SPEED!

LOW POWER!

- **Intel Celeron G4930**
  - 3.2 GHz
  - 2 Cores
  - 2 MB cache
  - Q2 2019
  - **PassMark: 2628**
  - TDP: 54 W

- **AMD Ryzen R1102G**
  - 1.2 GHz
  - 2 Cores
  - 4 MB cache
  - Q1 2020
  - PassMark: ~1600
  - **TDP: 6 W**

Motivation

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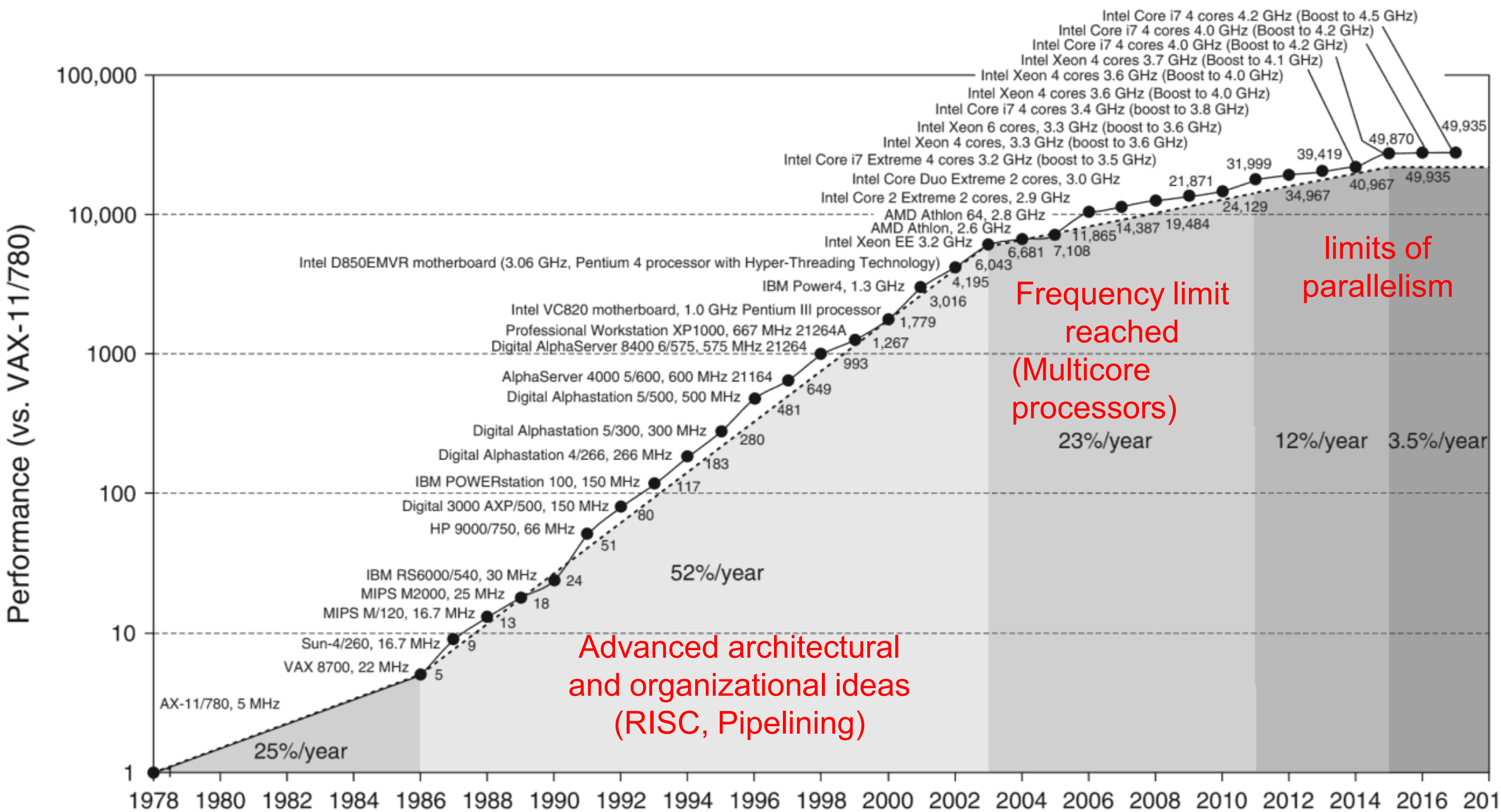
Growth in processor performance



Source: John L. Hennessy, David A. Patterson (2017). Computer Architecture: A Quantitative Approach. 6th edition.

Learning Objectives

- **At the end of this lesson you will be able**
  - to explain different types of bus architectures
  - to understand the difference between von Neumann and Harvard architecture
  - to understand RISC and CISC paradigms
  - to describe the idea of pipelining
  - to calculate processing performance improvement through pipelining
  - to describe the basics of parallel computing

Aspects of Optimization

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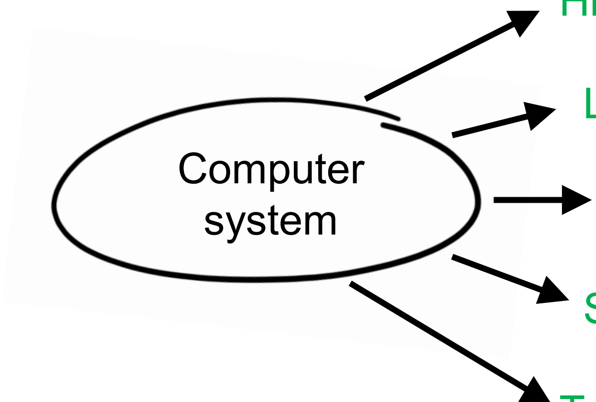
■ ‘Wishlist of Optimization’



Optimizing for:	**Drawbacks on:**
Higher speed	Power, cost, chip area
Lower cost	Speed, reliability
Zero power consumption	Speed, cost
Super reliable	Chip area, cost, speed
Temperature range	Power, cost, lifetime

- This lecture: Increasing system/CPU speed
- Some other topics: Covered in ‘Microcomputer Systems 1’ (MC1)

How to Increase System Speed?

Computer system performance

External factors:

- Better Compilers
- Better Algorithms

System level factors:

- Special Purpose Units
  e.g. Crypto, Video, AI
- Multiple Processors
- Bus Architecture
  e.g. von Neumann / Harward
- Faster components (e.g. SSDs, 1000Base-T, etc.)

CPU improvements:

- Clock Speed
- Cache Memory
- Multiple Cores / Threads
- Pipelined Execution
- Branch Prediction
- Out of Order Execution
- Instruction Set Architecture

Bus Architecture

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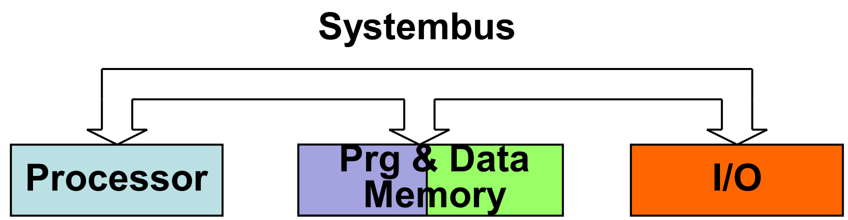
How is the connection between the CPU and memory organized?

A single system bus:	**simple, cheaper**	**slower**
Multiple system buses:	**faster**	**more complexity**

Bus Architecture

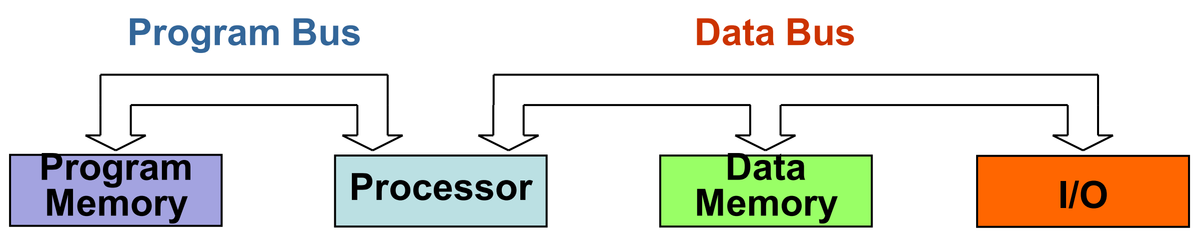
von Neumann Architecture

- Same memory holds program and data
- Single bus system between CPU and memory



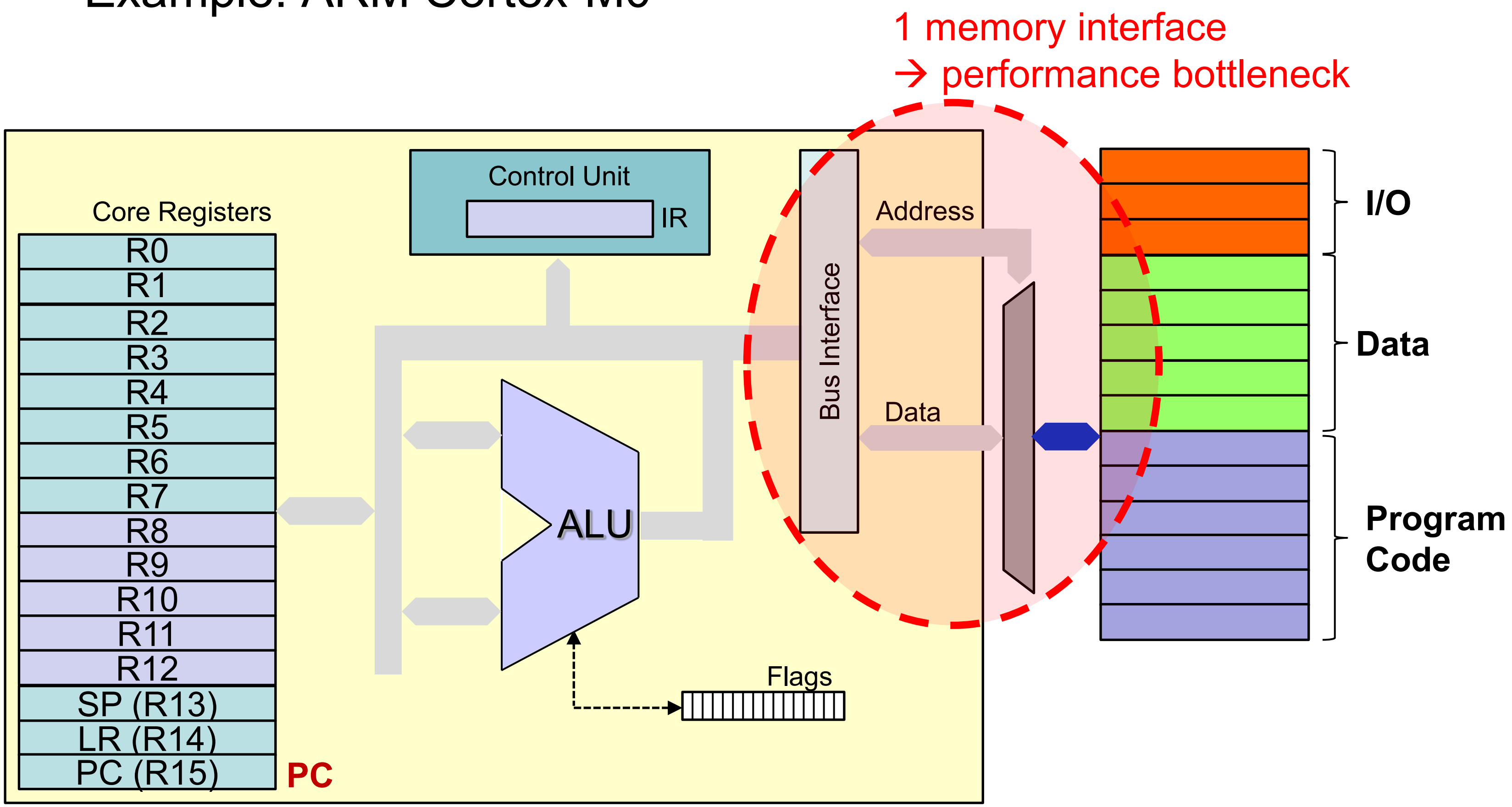
Harvard Architecture

- “Mark I” at Harvard University (Howard Aiken, 1939-44)
- Separate memories for program and data
- Two sets of address/data buses between CPU and memory



Bus Architecture

- **von Neumann Architecture**
  - Example: ARM Cortex-M0



Bus Architecture

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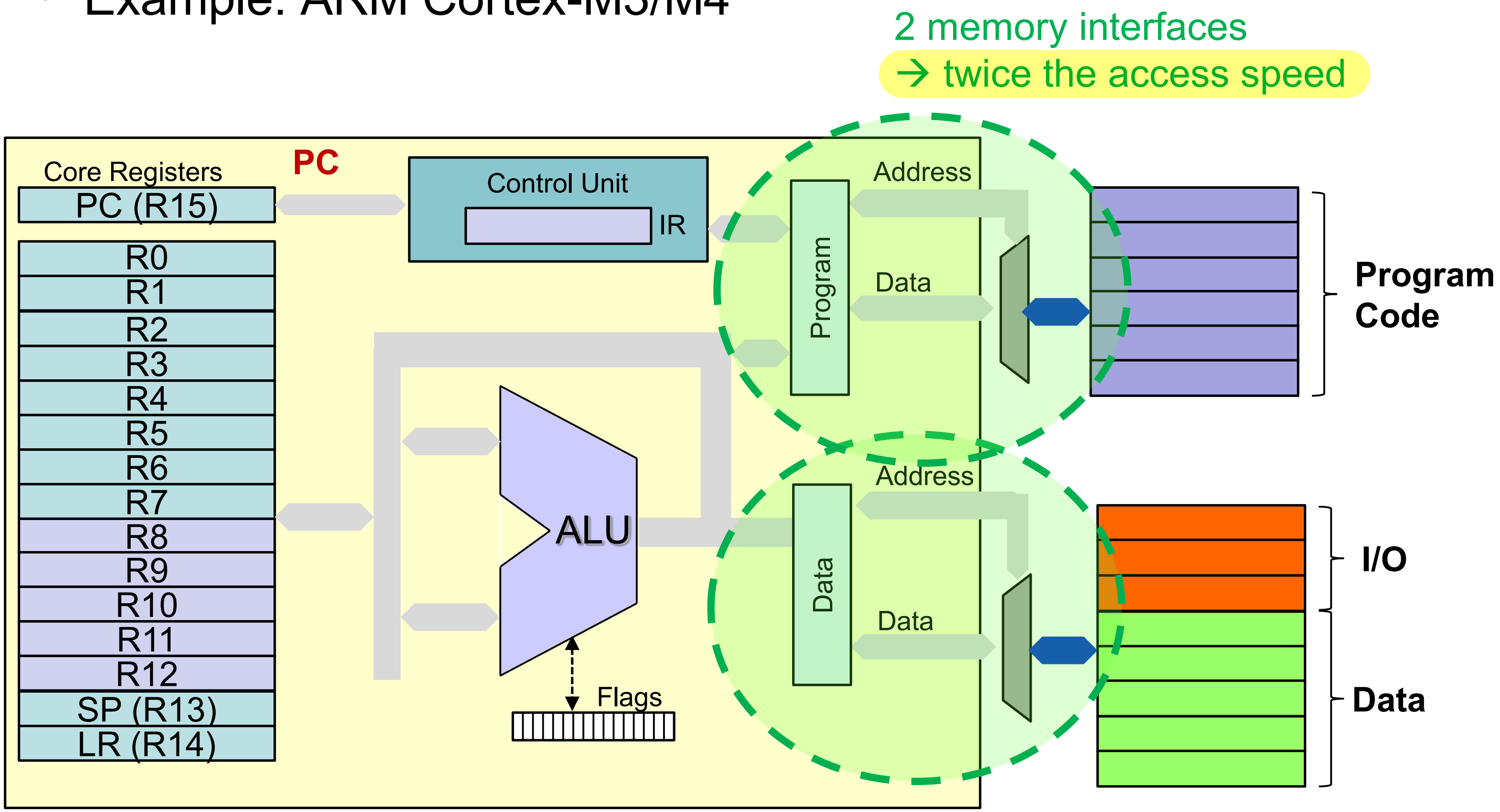
- **Harvard Architecture**
  - Example: ARM Cortex-M3/M4



Instruction Set Architecture

Powerful, complex instructions?

or

Simple, highly optimizable instructions?

(or something in between?)

Instruction Set Architecture

- **First CPUs**
  - Small address bus, small data bus (8 bit)
  - Few registers, no cache, pipelining, co-processors, multi-core
  - Machine code as finite state machine: Shift, multiply, etc. in loops

- **Consequences for machine code**
  - Differing complexity
  - Different length
  - Variable execution time
  - Individual addressing / arguments

  for different instructions

→ **More and more complex instructions (CISC)**

Instruction Set Architecture

- **John Cocke (IBM, 1974) proved that**
  - 80% of work was done using only 20% of the instructions
  - Most (complex) instructions were not used at all by compilers

- **RISC (Reduced Instruction Set Computer)**
  - Few instructions, unique instruction format
  - Fast decoding, simple addressing
  - less hardware → allows higher clock rates
  - more chip space for registers (up to 256!)
  - Load-store architecture reduces memory accesses, CPU works at full-speed on registers

Acorn Archimedes A3000
Source: Wikipedia Commons

Remark: Word "CISC" was introduced with "RISC"

Instruction Set Architecture

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Example: **Balance = Balance + Credit**

- **RISC**
  - Load / Store Architecture
  - Data processing instructions only available on registers

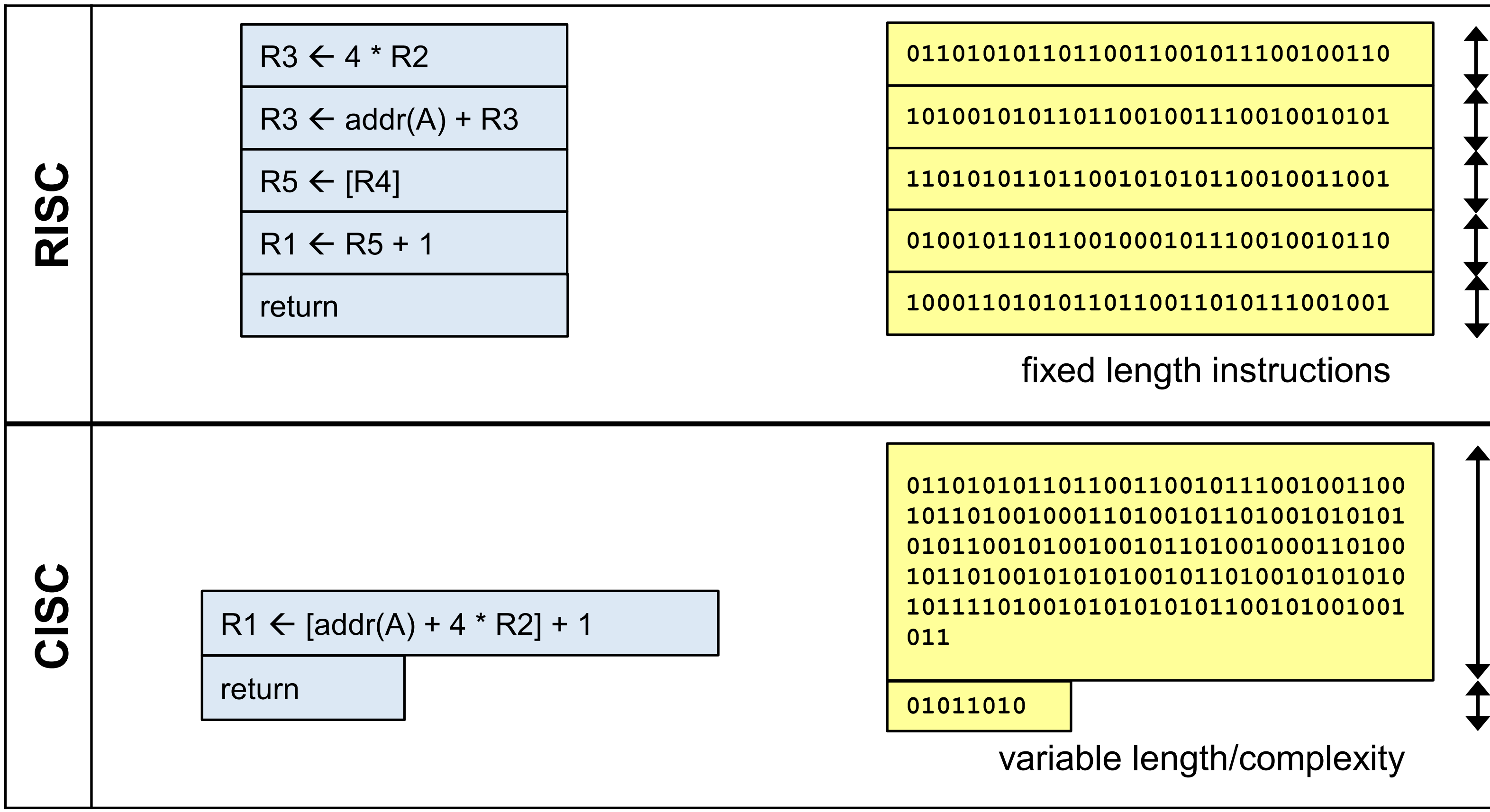
```
LDR   R0,=Credit
LDR   R1,[R0]
LDR   R0,=Balance
LDR   R3,[R0]
ADDS  R2,R1,R3
STR   R2,[R0]
```

- **CISC**
  - One of the operands of an instruction may directly be a memory location

```
MOV  AX, [Credit]
ADD  [Balance], AX
```

Instruction Set Architecture

- **Instruction / opcode complexity**



No RISC – No Fun

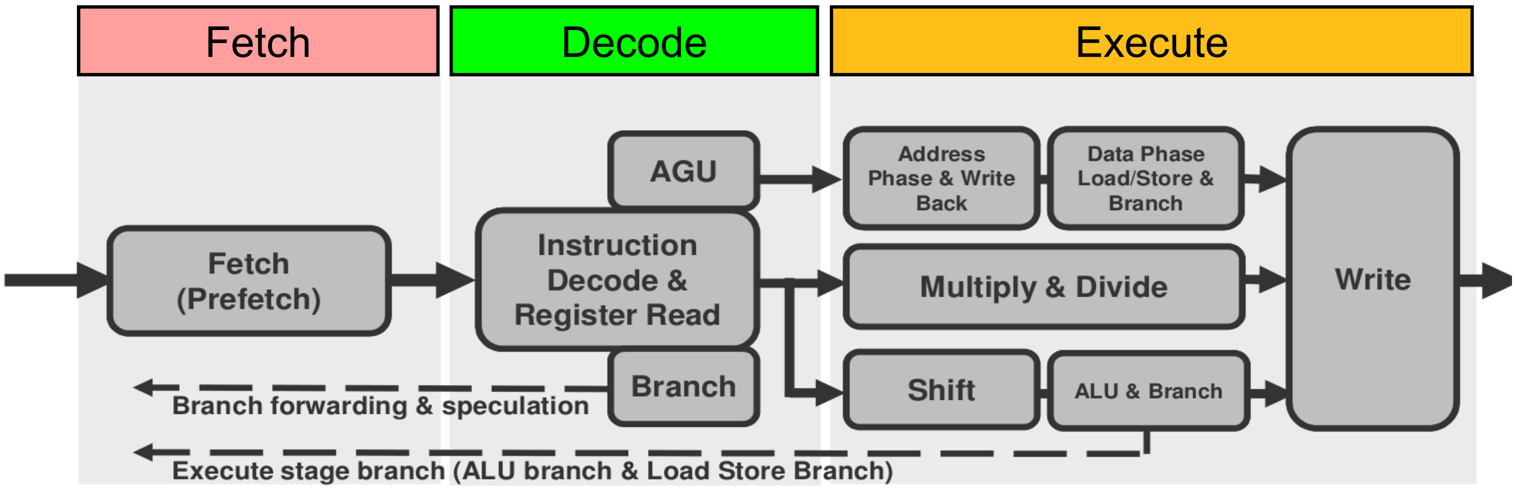
- **RISC advantages**
  - Simple and fast instruction decoding
  - More registers/cache on silicon area (less memory access)
  - Several data paths possible (superscalar computers)
  - Higher clock frequencies
  - Effective compiler optimization with limited, generic instructions
  - Easy pipelining, shorter pipelines (instruction size / duration)

- **CISC advantages**
  - Less program memory needed with complex instructions
  - Short programs may work faster with less memory accesses

- **CISC and RISC more and more indistinguishable**
  - Modern x86 CPUs convert instructions (CISC to RISC)

Pipelining

- **ARM Cortex-M3 sequence:**



Source: ARM

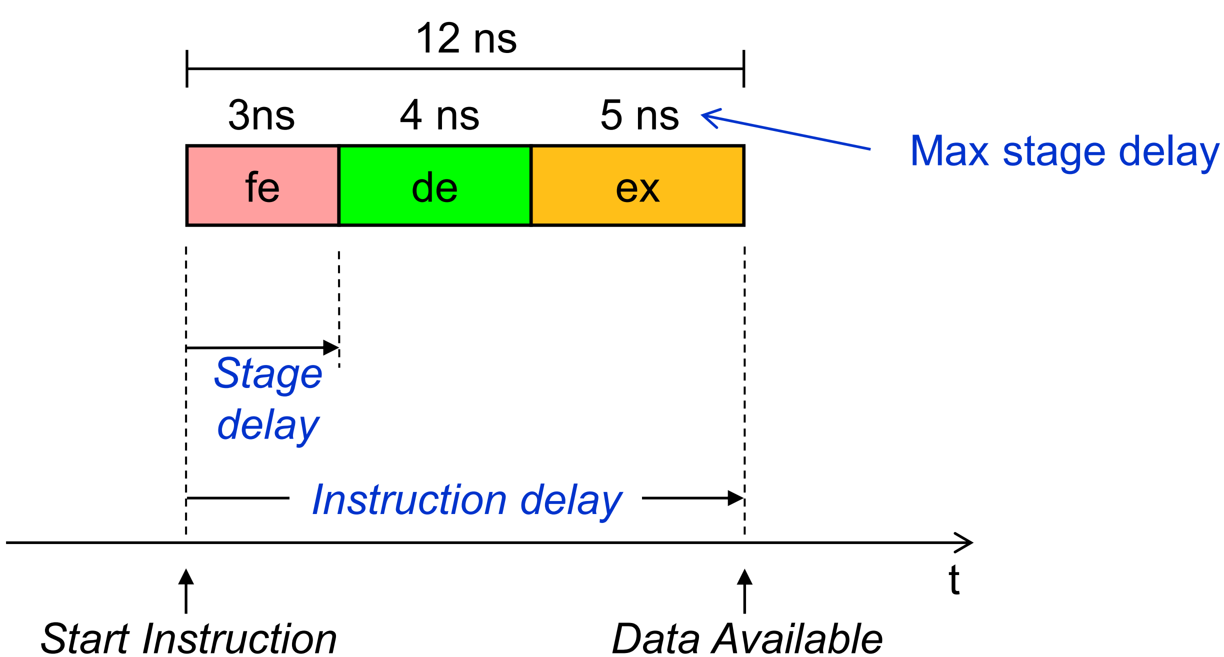
- **Idea: Why not fetch the next instruction, while the current one decodes?**

fe	de	ex		
	fe	de	ex	
		fe	de	ex

Pipelining

Timings and definitions

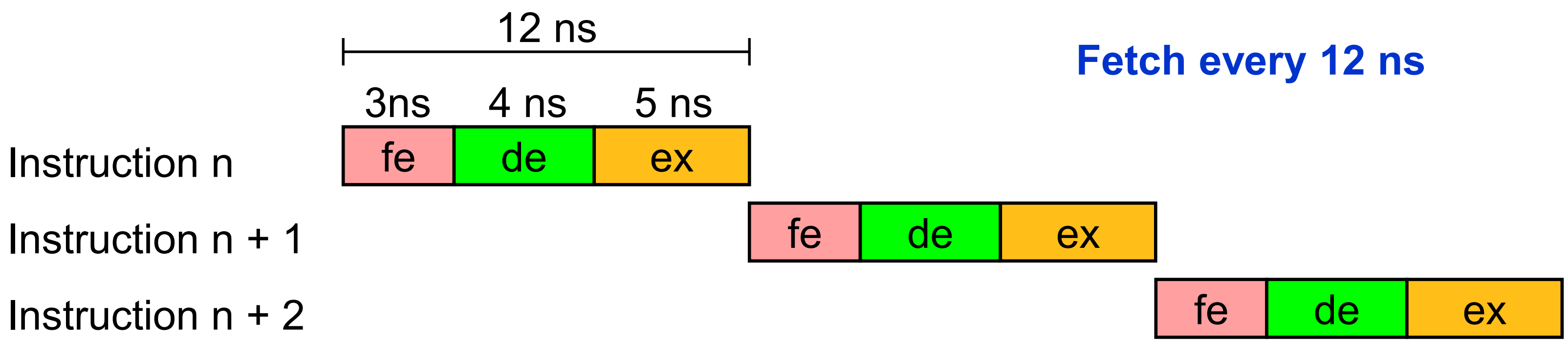
- Example: Cortex-M3
  - **fe: fetch** Read instruction 3 ns[1]
  - **de: decode** Decode instruction, read register or memory 4 ns[1]
  - **ex: execute** Execute instruction, write back result 5 ns[1]



[1]*timings estimated*

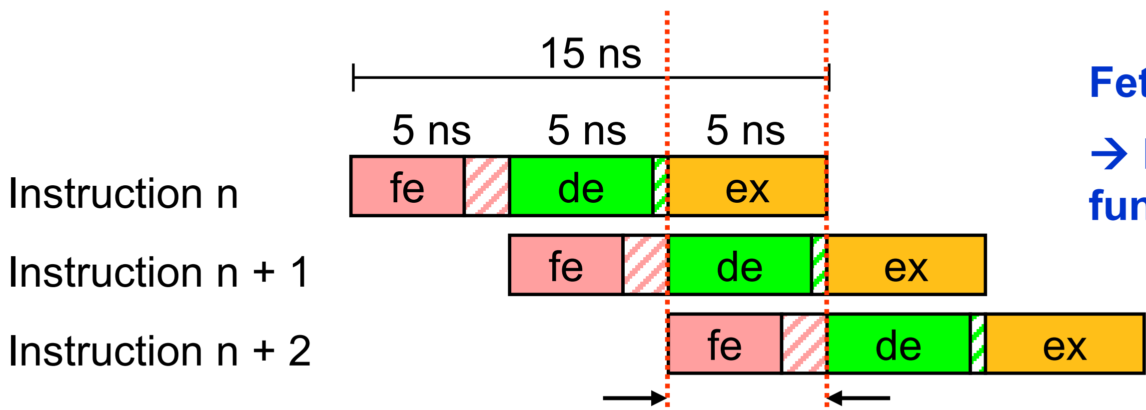
Pipelining

Sequential execution



Fetch every 12 ns

Pipelined execution



Longest stage (ex) defines pipeline cycle: 5 ns

Fetch every 5 ns

→ Better usage of functional units

Pipelining

Instructions per second

- Without pipelining

$$\text{Instructions per second} = \frac{1}{\text{Instruction delay}}$$

- With pipelining
  - Pipeline needs to be filled first
  - After filling, instructions are executed after every stage

$$\text{Instructions per second} = \frac{1}{\text{Max stage delay}}$$

Example Cortex-M3

- Without pipelining: 1/12ns = 83,3 million instructions per second
- With pipelining: 1/5ns = 200 million instructions per second

Pipelining

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- **Advantages**
  - All stages are set to the same execution time
  - Massive performance gain
  - Simpler hardware at each stage allows for a higher clock rate
- **Disadvantages**
  - A blocking stage blocks whole pipeline
  - Multiple stages may need to have access to the memory at the same time
- **General**
  - Number of execution stages is design decision
  - Typically 3 (ARM Cortex M4) – 20 stages (Pentium 4)

Pipelining

Optimal Pipelining

- All operations here are on registers (single cycle execution)
- In this example it takes 6 clock cycles to execute 6 instructions
- Clock cycles per instruction (CPI) = 1

Cycle			1	2	3	4	5	6	7	8	9
Operation											
ADD	fe	de	ex								
SUB		fe	de	ex							
ORR			fe	de	ex						
AND				fe	de	ex					
ORR					fe	de	ex				
EOR						fe	de	ex			

Pipeling

■ **Special situation: LDR**

- In this example it takes 7 clock cycles to execute 6 instructions
- Read cycle must complete on the bus before LDR instruction can complete
- Next 2 instructions must wait one pipeline cycle (S = stall)
- Clock cycles per instruction (CPI) = 1.2

Cycle			1	2	3	4	5	6	7	8	9
Operation											
ADD	fe	de	ex								
SUB		fe	de	ex							
LDR			fe	de	Ea	Ed					
AND				fe	de	S	ex				
ORR					fe	S	de	ex			
EOR							fe	de	ex		

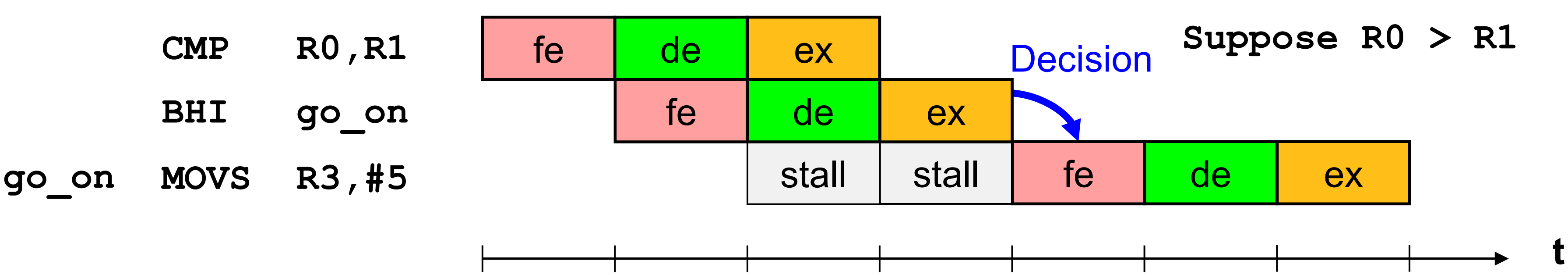
Ea: LDR address phase
Ed: LDR data phase

Pipelining

Control Hazards

- Branch / jump decisions occur in stage 3 (ex)
- Worst case scenario – conditional branch taken:

```
00000000 4288          CMP    R0,R1    ; R0 > R1 ?
00000002 D802          BHI    go_on    ; if higher -> go_on
00000004 000A          MOVS   R2,R1    ; otherwise exchange regs
00000006 0001          MOVS   R1,R0
00000008 0010          MOVS   R0,R2
0000000A 2305 go_on    MOVS   R3,#5
0000000C ...           ...
```



Pipeling

Reduce control hazards

- Loop fusion reduces control hazards
- Simple example for illustration

```
/* Two loops → double number of control hazards*/
for (i = 0; i < N; i++) {
        a[i] = 1/b[i] * c[i];
}
for (i = 0; i < N; i++) {
        d[i] = a[i] + c[i];
}

/* Better performance: Fusion into one loop */
for (i = 0; i < N; i++) {
        a[i] = 1/b[i] * c[i];
        d[i] = a[i] + c[i];
}
```

2*N Conditional Branches

N Conditional Branches

Pipelining

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- **Ideas to further improve pipelining:**
  - Branch prediction:
    - Store last decision made for each conditional branch
      → probability is high that the same decision is taken again
  - Instruction prefetch:
    - Fetch several instructions in advance
      → better use of the system bus
      → possibility of 'Out of Order Execution'
  - Out of Order Execution:
    - If one instruction stalls (e.g. a LDR from slow memory), it might be possible to already execute the next instruction

Pipeline

Limits of optimization

- Complex Optimizations → severe security problems (e.g. Out of Order Execution)
- Instructions (speculatively) executed, that would throw access violations under ‘In Order’ circumstances.
- ‘Meltdown’ and ‘Spectre’ attacks: allow a process to access the data of another process.

Parallel Computing

Streaming/Vector Processing

- One instruction processes multiple data items simultaneously

Multithreading

- Multiple programs/threads share a single CPU

Multicore Processors

- One processor contains multiple CPU cores

Multiprocessor Systems

- A computer system contains multiple processors

Parallel Computing

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- **Instructions working on more than one data item**

		Data streams	
		Single	Multiple
Instruction streams	Single	**SISD**: Single processor with one instruction, data stream	**SIMD**: Data Vectors, all data streams react to one instruction
	Multiple	(no examples)	**MIMD**: Multiprocessor with SIMD-instructions

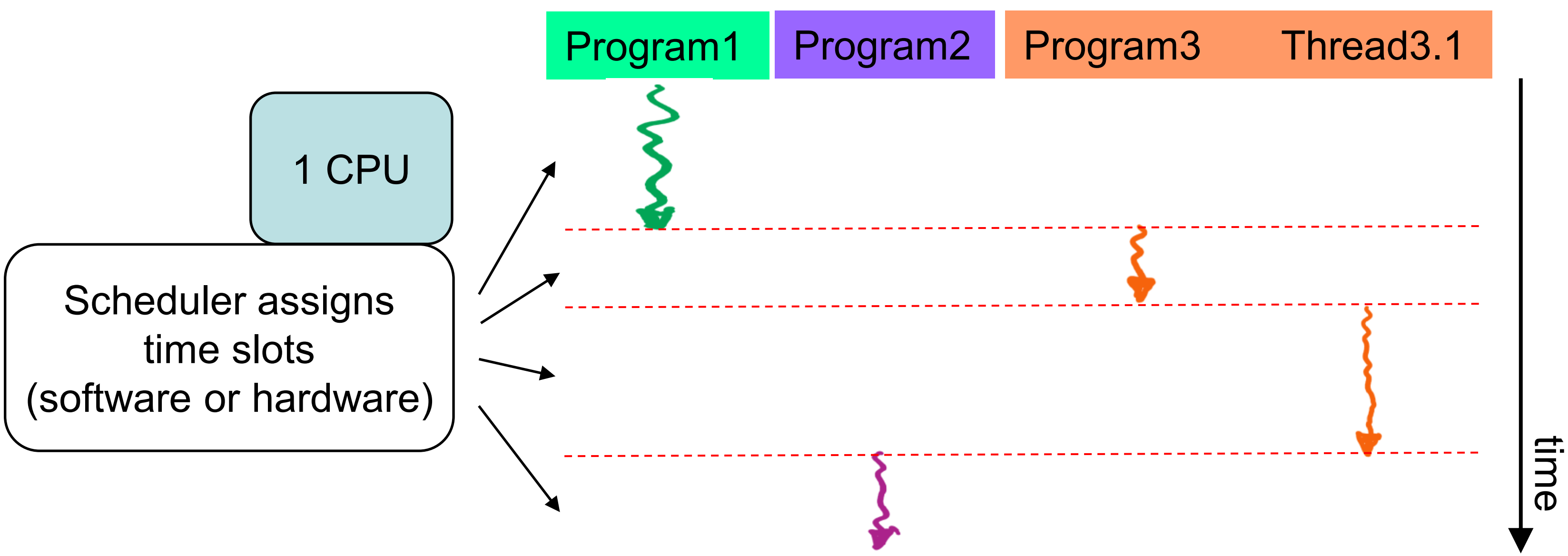
Source: David A. Patterson, John L. Hennessy

- **SIMD examples (x86):**

  MMX (Multimedia Extension), AMD 3D-Now,
  SSE (Streaming SIMD Extensions),
  AVX (Advanced Vector Extensions)

Parallel Computing

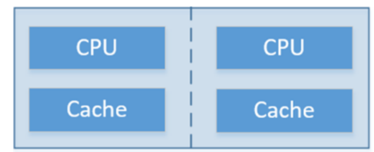
- **Multithreading**
  - Scheduler: Assigns time slots to programs/threads
  - Programs/threads only **seem** to run in parallel (1 CPU)



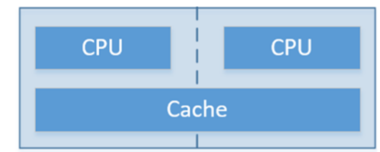
Remark: More on Multithreading in MC1

Parallel Computing

- **Parallelism on CPU / processor level:**
  - Multicore processor
    - All on one chip
    - Less traffic (cores integrated on one chip)
    - Possibility to share memories on-chip
    - Cheaper

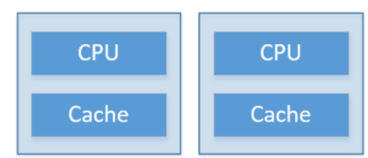


Multi-core



Multi-core with shared Cache

  - Multiprocessor
    - Multiple Chips
    - Longer distances between CPUs
    - More expensive



Multiprocessor

Source: https://www.queryhome.com/tech/106320/what-the-difference-between-multicore-and-multiprocessor

Conclusion

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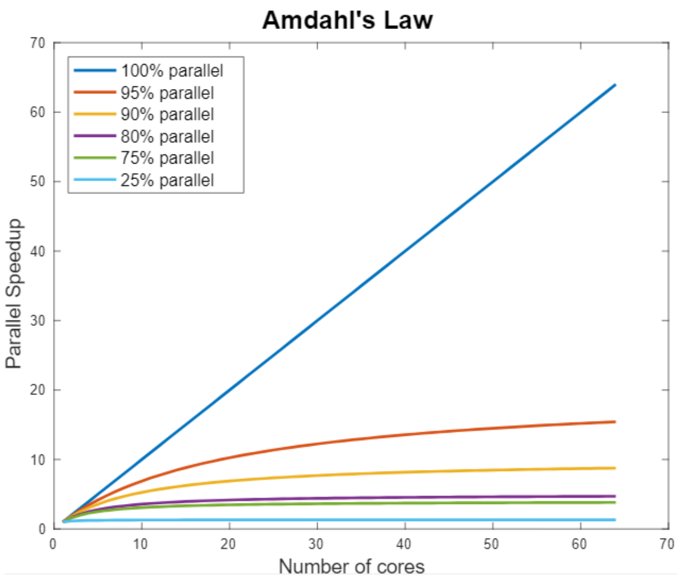
- **CPU clock frequency not the only factor**
- **Different ways to increase processor performance**
  - RISC / CISC
  - Pipelining
  - Out-of-order execution
  - Parallel computing
- **Today‘s CPUs combine most of these technologies**



Source:https://researchcomputingservices.github.io/parallel-computing/02-speedup-limitations/

- **Other components with influence**
  - SSD, memory, interfaces, algorithms …